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B.Tech. Degree IV Semester Special Supplementary Examination February 2020

ME 15 -1403 MECHATRONICS
(2015 Scheme)

Time : 3 Hours

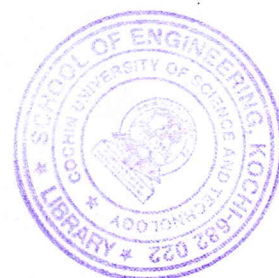
Maximum Marks : 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) A shaft encoder is used with a 50 mm radius tracking wheel to monitor linear displacement. If the encoder produces 256 pulses per revolution, what will be the number of pulses produced by a linear displacement of 200 mm?
- (b) For a thermocouple, what does the 'law of intermediate metals' say?
- (c) Explain the working of a piezoelectric sensor.
- (d) Describe the basic building blocks for modelling a thermal system.
- (e) Explain the action of a PID controller.
- (f) Comment on the location of roots of a system represented by the characteristic equation $9s^5 - 20s^4 + 10s^3 - s^2 - 9s - 10 = 0$.
- (g) Describe the working of a variable reluctance stepper motor.
- (h) Input A is applied directly to a two-input AND gate. Input B is applied to a NOT gate and then to the AND gate. What condition of inputs A and B will result in a 1 output?
- (i) What is the purpose of a Stack Pointer register?
- (j) When an input is given, an output is produced for 10 seconds and then it vanishes. Represent this using ladder diagram.



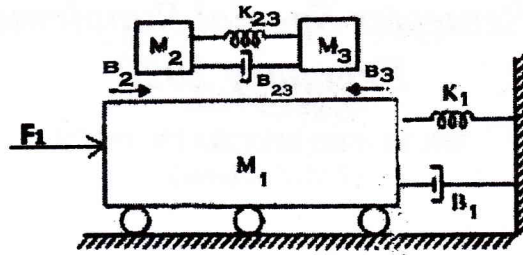
PART B

(4 × 10 = 40)

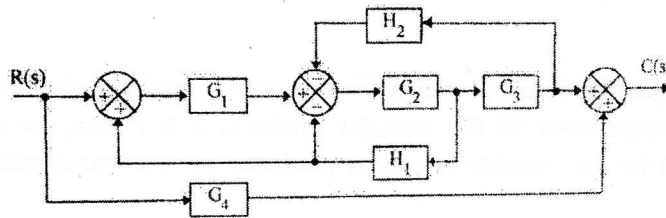
- II. (a) Describe bonded strain gauge and explain how it can be used to measure (i) torque (ii) gas pressure. (3)
 - (b) What are the different actuation methods of direction control valves? Draw the symbolic representation of each. (2)
 - (c) What is pulse width modulation? Draw a PWM signal with duty cycle 25%. Explain how the speed and direction of rotation of a D.C. motor can be controlled using a H-bridge circuit. (5)
- OR**
- III. (a) With a neat diagram, explain the working of rotary absolute encoder. For an absolute encoder with 14 tracks, what is the smallest angle that can be measured? (4)
 - (b) Draw and explain the characteristic of a thyristor. Describe the effect of increasing the gate current. (3)
 - (c) A double-acting pneumatic cylinder is controlled by a lever-actuated 5/3 DCV. Draw the circuit diagram and explain the action. (3)

(P.T.O.)

- IV. (a) Write the differential equations governing the mechanical system shown in the figure. Draw the force-voltage electrical analogous circuit and verify by writing the mesh equation. (5)

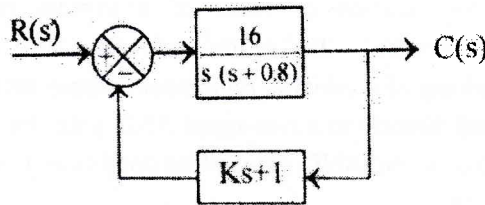


- (b) Obtain the closed loop transfer function $C(s)/R(s)$ of the system whose block diagram is shown in the figure. (5)



OR

- V. A position control system with velocity feedback is shown in the figure. What is the response $c(t)$ to the unit step input? Given that $\xi = 0.5$. Also calculate rise time, peak time, maximum overshoot and settling time. (10)



- VI. Sketch Bode plot for the following transfer function and determine the system gain K for the gain cross over frequency to be 5 rad/sec: (10)

$$G(s) = Ks^2 / [(1+0.2s)(1+0.02s)]$$

OR

- VII. Sketch the root locus of the system whose open loop transfer function is, $G(s) = K/[s(s+2)(s+4)]$. Find the value of K so that the damping ratio of the closed loop system is 0.5. (10)

- VIII. (a) Explain the working of an S-R flip-flop. (2)
- (b) A clock signal as a continuous sequence of pulses is applied to a logic gate and is to be given as output only when an enable signal is also applied to the gate. What logic gate can be used for this? (2)
- (c) Using a neat sketch showing the architecture, describe the components of a microprocessor. (6)

OR

- IX. (a) In a PLC-based automatic door system, the door is opened and closed by a double acting pneumatic cylinder. The cylinder is controlled using a 5/2 two-solenoid direction control valve. Draw the ladder diagram to achieve the following: when a push button is pressed the door opens; and when another push button is pressed the door closes. Draw also the pneumatic circuit diagram for DCV and cylinder. (5)
- (b) Explain the design and working of an automated system for operating car park barriers. (5)